Supplemental Material: Time-Dependent $c_s^2(a)$ from Ghost Condensation Dynamics

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We derive the time-dependent scalar perturbation sound speed $c_s^2(a)$ for the Blanchet–Skordis Khronon theory from first principles. The ghost condensation point $K'(Q_0{=}1) = 0$ is an attractor of the cosmological dynamics, so the field displacement $\delta(a) = Q(a) - 1$ decays as the universe expands. The associated sound speed $c_s^2 = \delta/(2{+}3\delta)$ is therefore time-dependent. However, we prove a no-go result: for the *pure* Khronon theory (without the BS2025 tensor field), no choice of kinetic function $K(Q)$ or running coupling $\mu(a)$ simultaneously yields (i) $c_s^2 \ll 1$ at recombination, (ii) $\Omega_{\rm DM} \simeq 0.265$ today, and (iii) CDM-like dilution $\rho_K \propto a^{-3}$. This clarifies the role of the Blanchet–Skordis (2025) tensor field as a necessary structural element for CMB compatibility.

I. INTRODUCTION

The constant DBI coupling $\alpha_{\rm DBI}$ used in the preceding analysis (Sec. VII of the main text) treats the background Khronon as fixed at the ghost condensation point $Q_0 = 1$. The matter power spectrum exclusion $\alpha_{\rm DBI} < 1.1 \times 10^{-7}$ (from $|\Delta P/P| < 5\%$ at $k = 0.2\ h/\mathrm{Mpc}$) then appears to contradict the CMB requirement $\alpha_{\rm DBI} \sim 0.01$ at recombination.

In this supplement we ask whether the *physical* time evolution of $Q_0(a)$ can resolve this tension. We derive $c_s^2(a)$ from the Khronon equation of motion, classify all possible scaling regimes, and establish a no-go theorem for the pure Khronon.

II. KHRONON EQUATION OF MOTION ON FRW

A. The conservation law

[KNOWN] On the FRW background with scale factor $a(t)$, the Blanchet–Skordis action [1] gives the Khronon field equation

$$\frac{d}{dt}\left[\frac{a^3\, K'(Q_0)}{N}\right] = 0, \tag{1}$$

where $Q_0 = \dot{\phi}/(Nc)$ is the background lapse ratio, N is the lapse function, and $K'(Q) = dK/dQ$. In cosmic time ($N = 1$):

$$a^3\, Q_0\, K'(Q_0) = I_0 = \text{const.} \tag{2}$$

B. Explicit $\delta(a)$ for the quadratic K

[DERIVED] For $K(Q) = \mu^2(Q{-}1)^2$ with constant μ, $K'(Q_0) = 2\mu^2\delta$ where $\delta = Q_0 - 1$. Equation (2) becomes

$$a^3\,(1{+}\delta)\,2\mu^2\delta = I_0. \tag{3}$$

Defining $\mathcal{C} = I_0/(2\mu^2) = \delta_0(1{+}\delta_0)$ (normalized at $a_0 = 1$), the displacement satisfies

$$\boxed{\delta(1{+}\delta) = \frac{\mathcal{C}}{a^3}.} \tag{4}$$

The positive root is

$$\delta(a) = \frac{-1 + \sqrt{1 + 4\mathcal{C}/a^3}}{2}. \tag{5}$$

Limits.—For $a \gg \mathcal{C}^{1/3}$ (late times, $\delta \ll 1$): $\delta \simeq \mathcal{C}/a^3 \propto a^{-3}$. For $a \ll \mathcal{C}^{1/3}$ (early times, $\delta \gg 1$): $\delta \simeq \sqrt{\mathcal{C}}/a^{3/2} \propto a^{-3/2}$.

C. Ghost condensation as attractor

Lemma 1 (Attractor property)**.** *For any kinetic function $K(Q)$ with a minimum at $Q_0^* = 1$ (i.e., $K'(1) = 0$, $K''(1) > 0$), the cosmological solution $\delta(a) = Q_0(a) - 1$ satisfies $\delta(a) \to 0$ as $a \to \infty$.*

Proof. Define $f(\delta) \equiv (1{+}\delta)\,K'(1{+}\delta)$. Near $\delta = 0$: $f(\delta) \simeq K''(1)\,\delta > 0$ for $\delta > 0$. The conservation law $a^3 f(\delta) = I_0$ requires $f(\delta) = I_0/a^3 \to 0$. Since f is continuous with $f(0) = 0$ and $f'(0) = K''(1) > 0$, the unique solution is $\delta \to 0$. □

Physical meaning: The ghost condensation vacuum $Q = 1$ is a cosmological attractor. Regardless of the initial displacement, the Khronon field relaxes toward $Q = 1$ as the universe expands. The rate of approach depends on the form of $K(Q)$.

III. BACKGROUND SOUND SPEED

A. The k-essence formula

[KNOWN] For a k-essence scalar with kinetic function $K(Q)$, the adiabatic perturbation sound speed is

$$c_s^2 = \frac{K'(Q_0)}{K'(Q_0) + 2\,Q_0\,K''(Q_0)}, \tag{6}$$

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evaluated at the background $Q_0 = 1 + \delta$, *not* at the condensation minimum $Q = 1$ [4, 5].

B. Quadratic K: explicit $c_s^2(\delta)$

[DERIVED] For $K = \mu^2(Q-1)^2$:

$$c_s^2 = \frac{\delta}{2 + 3\delta}. \tag{7}$$

Limits: $c_s^2 \to \delta/2$ as $\delta \to 0$; $c_s^2 \to 1/3$ as $\delta \to \infty$ (stiff matter).

C. DBI K: explicit $c_s^2(\delta)$

[DERIVED] For $K_{\rm DBI} = 2\mu^2\lambda_D^2[\sqrt{1+\delta^2/\lambda_D^2} - 1]$, defining $R = \sqrt{1+\delta^2/\lambda_D^2}$:

$$c_s^2 = \frac{\delta\, R^2}{\delta\, R^2 + 2\,(1{+}\delta)\,\lambda_D^2}. \tag{8}$$

Limits: $c_s^2 \to \delta/(2\lambda_D^2)$ as $\delta \to 0$; $c_s^2 \to \delta^2/(\delta^2 + 2\lambda_D^4)$ for $\delta \gg \lambda_D$; $c_s^2 \to 1$ for $\delta \gg \lambda_D^2$.

D. General K: saturation theorem

Theorem 2 (Sound speed saturation)**.** *For any kinetic function $K(Q)$ satisfying $K(Q) \sim \mu^2\,|Q{-}1|^\alpha$ for $|Q{-}1| \gg 1$ with $\alpha \geq 1$:*

$$\lim_{\delta\to\infty} c_s^2 = \frac{1}{2\alpha - 1}. \tag{9}$$

In particular: $c_s^2 \to 1/3$ for $\alpha = 2$ (quadratic); $c_s^2 \to 1$ for $\alpha = 1$ (DBI linear). For all $\alpha \geq 1$, $c_s^2 = O(1)$ whenever $\delta \gg 1$.

Proof. For $K \sim \mu^2\delta^\alpha$: $K' \sim \alpha\mu^2\delta^{\alpha-1}$, $K'' \sim \alpha(\alpha{-}1)\mu^2\delta^{\alpha-2}$. In the limit $\delta \gg 1$ (so $Q_0 \simeq \delta$), Eq. (6) gives

$$\begin{aligned} c_s^2 &\simeq \frac{\alpha\delta^{\alpha-1}}{\alpha\delta^{\alpha-1} + 2\delta\cdot\alpha(\alpha{-}1)\delta^{\alpha-2}} \\ &= \frac{1}{1 + 2(\alpha{-}1)} = \frac{1}{2\alpha{-}1}. \end{aligned} \tag{10}$$

For $\alpha = 2$: $c_s^2 = 1/(2\cdot 2 - 1) = 1/3$. For $\alpha = 1$: $c_s^2 = 1/(2-1) = 1$. □

IV. TIME EVOLUTION: $c_s^2(a)$

A. Quadratic K, constant μ

[PROVEN] Combining Eqs. (5) and (7):

$$c_s^2(a) = \frac{\delta(a)}{2 + 3\,\delta(a)}, \qquad \delta(a) = \frac{-1 + \sqrt{1 + 4\mathcal{C}/a^3}}{2}. \tag{11}$$

Late-time scaling ($a \gg 1$, $\delta \ll 1$):

$$c_s^2(a) \simeq \frac{\delta_0}{2\,a^3} \propto a^{-3}. \tag{12}$$

Early-time saturation ($a \ll 1$, $\delta \gg 1$):

$$c_s^2(a) \to \frac{1}{3}. \tag{13}$$

Table I shows the full evolution.

B. Quadratic K, running $\mu = H/c$

[PROVEN] With $\mu(a) = H(a)/c$, the conservation law gives

$$\delta(a) = \frac{I_0\, c^2}{2\, H(a)^2\, a^3}. \tag{14}$$

In the matter era ($H^2 \propto a^{-3}$), δ is *constant*:

$$c_s^2 = \frac{\delta_{\rm mat}}{2 + 3\,\delta_{\rm mat}} \simeq 0.1\text{–}0.2 = \text{const.} \tag{15}$$

This is excluded by TKS [6] ($c_s^2 < 3.2\times10^{-6}$ at 95% CL).

V. NO-GO THEOREM

Theorem 3 (No-go for pure Khronon)**.** *Consider the pure Khronon theory with any analytic kinetic function $K(Q)$ satisfying:* (i) *ghost condensation at $Q = 1$: $K'(1) = 0$, $K''(1) > 0$;* (ii) *positive energy: $QK' - K > 0$ at the background. Then there exists no function $\mu(a)$ such that simultaneously:*

1. $\Omega_{\rm DM}(a_0) = 0.265$ *(dark matter density today),*
2. $c_s^2(a_{\rm rec}) < 10^{-6}$ *(CMB compatibility),*
3. $\rho_K \propto a^{-3}$ *to within 1% from matter-radiation equality to today.*

TABLE I. Evolution of $\delta(a)$ and $c_s^2(a)$ for $K = \mu^2(Q{-}1)^2$, constant μ, $\delta_0 = 0.34$. The sound speed saturates at 1/3 for $\delta \gg 1$ (early times).

a	z	$\delta(a)$	$c_s^2(a)$
1	0	0.34	0.113
10^{-1}	9	21	0.319
10^{-2}	99	675	0.333
10^{-3}	999	2.1×10^4	0.333
10	—	2.3×10^{-4}	1.1×10^{-4}
10^2	—	2.3×10^{-7}	1.1×10^{-7}

Proof. By the conservation law, $\delta(a)$ is determined by $\mu(a)$ through Eq. (2). We consider three regimes for $\delta(a_{\rm rec})$.

Case 1: $\delta(a_{\rm rec}) \ll 1$.—Then $c_{\rm s}^2(a_{\rm rec}) \simeq \delta(a_{\rm rec})/2$, so condition (2) requires $\delta(a_{\rm rec}) < 2\times 10^{-6}$. We now show that condition (3) is violated for *any* $\mu(a)$ achieving this bound, by an exact, model-independent calculation.

From the conservation law $\mu^2\delta = I_0/(2a^3)$, the energy density (Eq. (**??**) of the main text) takes the form

$$\rho_K(a) = \frac{c^2 I_0}{16\pi G}\,\frac{2+\delta(a)}{a^3}. \tag{16}$$

For exact CDM scaling ($\rho \propto a^{-3}$), the factor $(2+\delta)$ must be constant. Normalizing to $a_0 = 1$, the fractional deviation from a^{-3} at any epoch a is

$$\Delta(a) \;\equiv\; \frac{\rho_K(a)}{\rho_K(a_0)\,(a_0/a)^3} - 1 \;=\; \frac{2+\delta(a)}{2+\delta_0} - 1. \tag{17}$$

This expression is exact and depends on $\mu(a)$ only through $\delta(a)$.

Now evaluate at $a = a_{\rm rec}$. Since $0 \le \delta(a_{\rm rec}) < 2\times 10^{-6}$:

$$\Delta(a_{\rm rec}) = \frac{2+\delta(a_{\rm rec})}{2+\delta_0} - 1 \;\in\; \left[\frac{2}{2+\delta_0} - 1,\;\; \frac{2+2\times 10^{-6}}{2+\delta_0} - 1\right]. \tag{18}$$

With $\delta_0 = 0.34$ (fixed by condition (1)):

$$\Delta(a_{\rm rec}) = \frac{2}{2.34} - 1 + O(10^{-6}) = -\frac{\delta_0}{2+\delta_0} + O(10^{-6}) = -0.1453 \tag{19}$$

That is, the energy density at recombination is 14.5% *below* the a^{-3} extrapolation from today—regardless of how $\mu(a)$ varies. This exceeds the 1% tolerance of condition (3) by more than an order of magnitude.

Model-independence.—The bound (19) uses only two ingredients: (a) the identity $\rho_K \propto (2+\delta)/a^3$, which follows from the conservation law and is independent of $\mu(a)$; and (b) the boundary values $\delta_0 = 0.34$ (condition 1) and $\delta_{\rm rec} < 2\times 10^{-6}$ (condition 2 in this regime). No assumption about the functional form of $\mu(a)$ or the intermediate trajectory $\delta(a)$ is needed. The deviation $|\Delta| = \delta_0/(2+\delta_0)$ coincides with the equation of state $w_0 = \delta_0/(2+\delta_0) = 0.145$, which is the irreducible cost of matching $\Omega_{\rm DM}$ with a non-constant $(2+\delta)$ factor. This completes Case 1.

Case 2: $\delta(a_{\rm rec}) \gg 1$.—By Theorem 2, $c_{\rm s}^2 = O(1)$, violating condition (2).

Case 3: $\delta(a_{\rm rec}) \sim O(1)$.—Then $c_{\rm s}^2 \sim 0.1$–0.3 (from Eq. 7), again violating condition (2) by $\gtrsim 5$ orders of magnitude. □

Completeness.—The three cases are exhaustive because $\delta(a_{\rm rec})$ is a non-negative real number. Case 1 ($\delta \ll 1$), Case 2 ($\delta \gg 1$), and Case 3 ($\delta \sim O(1)$) cover the entire range. No fine-tuned $\mu(a)$ can avoid this trichotomy: regardless of the path $\mu(a)$ takes, the value $\delta(a_{\rm rec})$ must fall in one of these three regimes, and each violates at least one condition.

Remark.—The theorem applies to the *pure* Khronon and to any DBI completion (8). It does *not* apply to theories with additional fields (such as the BS2025 tensor field [2]), which modify the perturbation structure beyond the k-essence formula.

VI. IMPLICATIONS

A. The BS2025 tensor field as structural necessity

The no-go theorem (Theorem 3) elevates the Blanchet–Skordis tensor field [2] from a *convenience* ("one way to achieve $c_{\rm s}^2 = 0$") to a *structural necessity* for the Khronon dark matter program. The pure Khronon with any $K(Q)$ cannot simultaneously reproduce the CMB and the correct dark matter abundance. The tensor field provides the additional degree of freedom needed to decouple the background equation of state from the perturbation sound speed.

B. Reinterpretation of $\alpha_{\rm DBI}$

Within the BS2025 framework (where the tensor field enforces $c_{\rm s}^2 = 0$ at leading order), the DBI parameter $\alpha_{\rm DBI} = Q_0/(2\lambda_D^2)$ represents a *next-to-leading-order* correction that is suppressed by the tensor field dynamics. The P(k) constraint $\alpha_{\rm DBI} < 1.1\times 10^{-7}$ then translates to a bound on the tensor field mass or coupling, not directly on λ_D.

C. What the time dependence teaches us

Although the time dependence of $c_{\rm s}^2(a)$ does not resolve the CMB crisis for the pure Khronon, it reveals the correct physical picture:

1. The ghost condensation point is a cosmological *attractor* (Lemma 1): the Khronon field rolls toward $Q = 1$ as the universe expands.
2. The approach rate is $\delta \propto a^{-3}$ for the quadratic K with constant μ, giving $c_{\rm s}^2 \propto a^{-3}$ at late times.
3. The saturation $c_{\rm s}^2 \to 1/3$ (quadratic) or $c_{\rm s}^2 \to 1$ (DBI) at early times when $\delta \gg 1$ is a *generic* feature of displaced ghost condensates.
4. The correct CLASS implementation should use the time-dependent $c_{\rm s}^2(a)$ from Eq. (11), not a constant $\alpha_{\rm DBI}$.

VII. SUMMARY OF FORMULAS

For reference, we collect the key formulas for CLASS implementation.

Quadratic K, constant μ:

$$\delta(a) = \frac{-1+\sqrt{1+4\mathcal{C}/a^3}}{2}, \quad \mathcal{C} = \delta_0(1+\delta_0), \tag{20}$$

$$c_{\rm s}^2(a) = \frac{\delta(a)}{2+3\,\delta(a)}, \tag{21}$$

$$w(a) = \frac{\delta(a)}{2+\delta(a)}. \tag{22}$$

DBI K, constant μ: $\delta(a)$ from

$$\frac{(1+\delta)\,\delta}{\sqrt{1+\delta^2/\lambda_D^2}} = \frac{\mathcal{C}_{\rm DBI}}{a^3}, \tag{23}$$

with $c_{\rm s}^2$ from Eq. (8).

VIII. CONCLUSION

The scalar perturbation sound speed $c_{\rm s}^2$ in the Blanchet–Skordis Khronon theory is *not* constant. It evolves with the scale factor through the field displacement $\delta(a)$, which decays as the ghost condensation attractor pulls $Q_0 \to 1$. The exact time dependence is given by Eqs. (20)–(21) for the quadratic kinetic function.

However, this time dependence does *not* resolve the fundamental tension between CMB compatibility ($c_{\rm s}^2 < 10^{-6}$) and the observed dark matter abundance ($\Omega_{\rm DM} = 0.265$). Theorem 3 establishes that no pure Khronon theory with any $K(Q)$ and any $\mu(a)$ can satisfy both constraints simultaneously. The Blanchet–Skordis (2025) tensor field—or an equivalent additional degree of freedom—is a *structural necessity* for the Khronon dark matter program, not merely a theoretical preference.

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